

SRI RAM MANNAM

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EDUCATION

San Jose State University

San Jose, CA

Master of Science in Software Engineering

May 2026

Relevant Coursework: Enterprise Distributed Technologies, Engineering Software Systems, Software Defined Networks, Virtualization Technologies, Machine Learning, Mobile Application Development

Prasad V. Potluri Siddhartha Institute of Technology

Vijayawada, India

Bachelor of Technology in Information Technology

May 2024

PROJECTS

Rideshare Application – Distributed Microservices Backend ([demo](#) · [code](#)) *TypeScript, Node.js/Express, Python (FastAPI), PostgreSQL, Redis, Docker, GKE, GitHub Actions, Linux*

- Broke a monolithic prototype into 11 independently deployable Node.js/Express services plus a Python FastAPI trip-grouping service, working with a small Agile team so the system kept serving rides even when single services failed.
- Built REST APIs and tuned PostgreSQL schemas, indexes, and queries, then added Redis caching on the hottest trip and user endpoints and JWT/RBAC guards, cutting p95 response times on read-heavy routes and locking down rider, driver, and payment data.
- Containerized services with Docker and shipped them to GKE, wiring GitHub Actions pipelines that lint, test, and roll out images on each merge; used logs and basic metrics from live pods to track down slow queries and misconfigured services.

PIC Standard (Provenance & Intent Contracts) – Open Source Contributor ([code](#)) *TypeScript, Python, Docker, HTTP, Observability, Security*

- Put together a Dockerized HTTP bridge for agentic provenance services with Python health checks and slimmer image builds, making it easier for contributors to run the stack locally and keep it alive in long-running environments.
- Tightened observability and security by adding audit-logging middleware, cleaning up X-Request-ID tracing, and exposing version endpoints in the PIC SDK so downstream services can see which build handled a request when something fails.

Restaurant Finder Application ([demo](#) · [code](#))

Python, Flask, PostgreSQL, pytest, AWS EC2, AWS RDS

- Owned the Flask backend that powers restaurant search and reviews, defining the REST API surface with teammates so the frontend could ship new flows without constantly waiting on backend changes.
- Reworked PostgreSQL schemas and added the right indexes on search and review tables, trimming query time by roughly 50% on the main search path and making user-facing pages feel noticeably snappier on the EC2/RDS stack.
- Wired GitHub Actions to run pytest against core API and DB logic on every push and deploy green builds to AWS, which caught breaking changes before they hit EC2 and gave the team a single view of which commit was live.

DocSense AI – Agentic RAG PDF Reader ([demo](#) · [code](#)) *Python, Flask, Qdrant, FastEmbed, Redis, Supabase, Docker*

- Built an agentic RAG backend for PDFs that decomposes queries and runs hybrid retrieval (FastEmbed dense vectors + BM25 + Reciprocal Rank Fusion), keeping follow-up questions grounded in the right paragraphs with citations.
- Split the system across a lean Flask orchestration app, dedicated embedding and reranking services, and a GitHub Actions ingestion runner, adding a two-tier cache (TTL + Redis/Upstash), Supabase JWT auth, and Groq Llama 3.3 70B fallback before deploying to Render.

ADDITIONAL EXPERIENCE

Wells Fargo Software Engineering Virtual Experience – Forage ([cert](#))

Feb 2026

- Modeled a banking-style ERD in Java, translating accounts, customers, and transactions into relational entities and implementing them with OOP so the schema and code stayed in sync; published the solution to GitHub.

Walmart USA Advanced Software Engineering Virtual Experience – Forage ([cert](#))

Aug 2026

- Built a custom d-ary heap in Java and designed ERD/UML models while wrangling noisy CSV and database data into a single, standardized schema through data wrangling (data munging).

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL; actively learning Go; strong foundation in data structures and algorithms

Databases: PostgreSQL, MySQL (schema design, indexing, query optimization); basic experience with DynamoDB, MongoDB

Backend Frameworks & Tools: Flask, Spring Boot, Node.js/Express, gRPC, RESTful APIs, WebSockets, pytest

CI/CD & DevOps: GitHub Actions, Docker, Kubernetes (GKE, manifests and higher-level tooling), CI/CD pipelines

Version Control & Agile: Git, JIRA, Agile/Scrum, code reviews, collaborative multi-person projects and pair programming

OS & Infrastructure: Linux (Ubuntu, command-line administration), AWS (EC2, RDS, S3, Lambda), GCP

Core Competencies: Distributed systems, API design, scalable backend development, automated testing, observability, system design, end-to-end ownership of services

Certifications: [GitHub Copilot](#), [AWS-Cloud Practitioner](#), [Anthropic-Claude Code 101](#)

LEADERSHIP EXPERIENCE

SmartInterviews – Student Mentor ([cert](#))

May 2023–Mar 2024

- Ran 50+ code reviews and sprint-style practice cycles with 100+ students on data structures, algorithms, and backend topics, giving targeted feedback on REST APIs, OOP design, and Git hygiene.
- Wrote and presented a small curriculum on backend development, RESTful APIs, and scalable system design, turning abstract topics into worked examples and live walk-throughs that beginners could actually ship and debug.